# Hyunsik Jeon

AI & Backend Developer Portfolio

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## FEATURED PROJECT 01

**Top Award — IBM & RedHat AI Transformation Program**

### Vaccine Daily (AI News Pipeline)

Hyunsik Jeon and 2 others

*React · FastAPI · PostgreSQL · ChromaDB · GPT-4o-mini · ko-sroberta · HDBSCAN · GraphRAG · Docker*

An automated news intelligence pipeline that crawls 10 major media outlets every 5 minutes and uses a multi-agent workflow to generate verified, hallucination-free personalized reports.

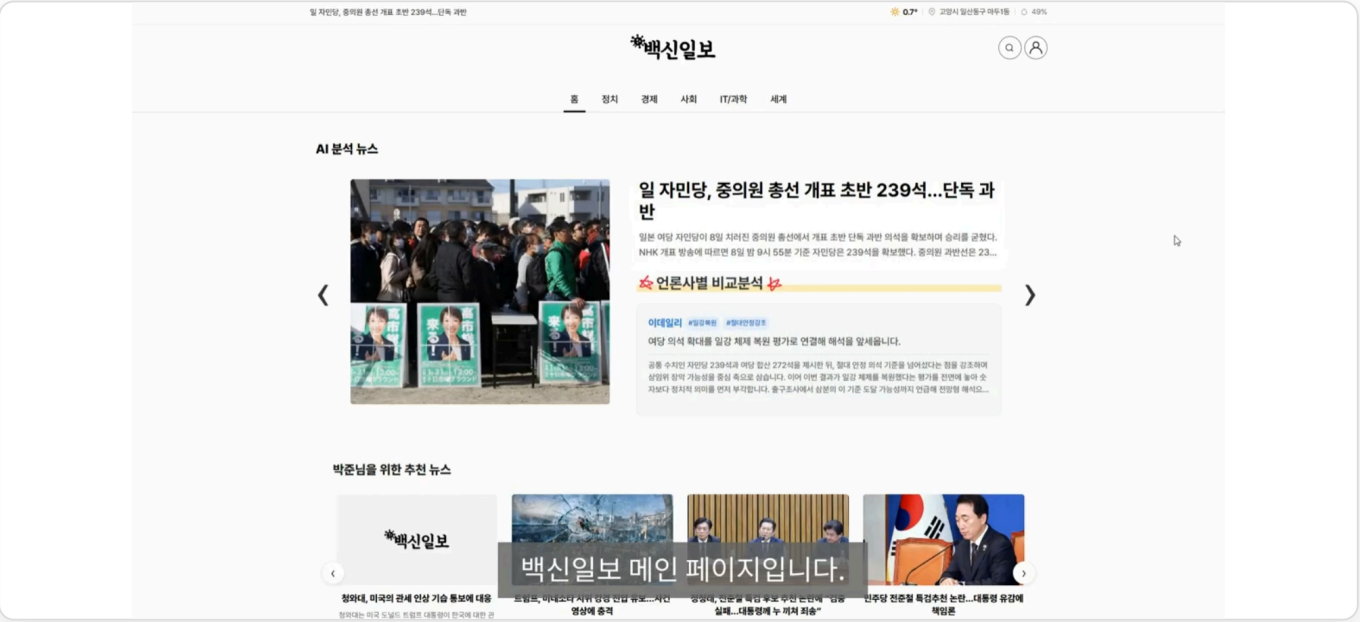
**Technical Pipeline Detail**

**1. Crawling** Real-time automated data collection from 10 media outlets via Selenium & BS4.

**2. Clustering** Token-efficient group analysis of similar news using HDBSCAN and ko-sroberta.

**3. AI News Article Generator** Multi-Agent AI (Writer → Critic → Editor) for factual verification and summarization.

**4. GraphRAG** Entity-Relation extraction and Knowledge Graph construction for Media Tone of Voice analysis.



## FEATURED PROJECT 02

# Football AI Prediction System

Hyunsik Jeon (Personal Use)

*Python · TensorFlow · Reinforcement Learning · Monte Carlo · XGBoost · LightGBM · FastAPI · Pandas*

A high-precision forecasting system analyzing data from 13 European football leagues. It maximizes predictive accuracy by ensembling deep learning models with reinforcement learning strategies.

### 6-Step System Architecture

**Step 1** Data Scrapping: Automated pipeline for match data and global bookmaker odds collection.

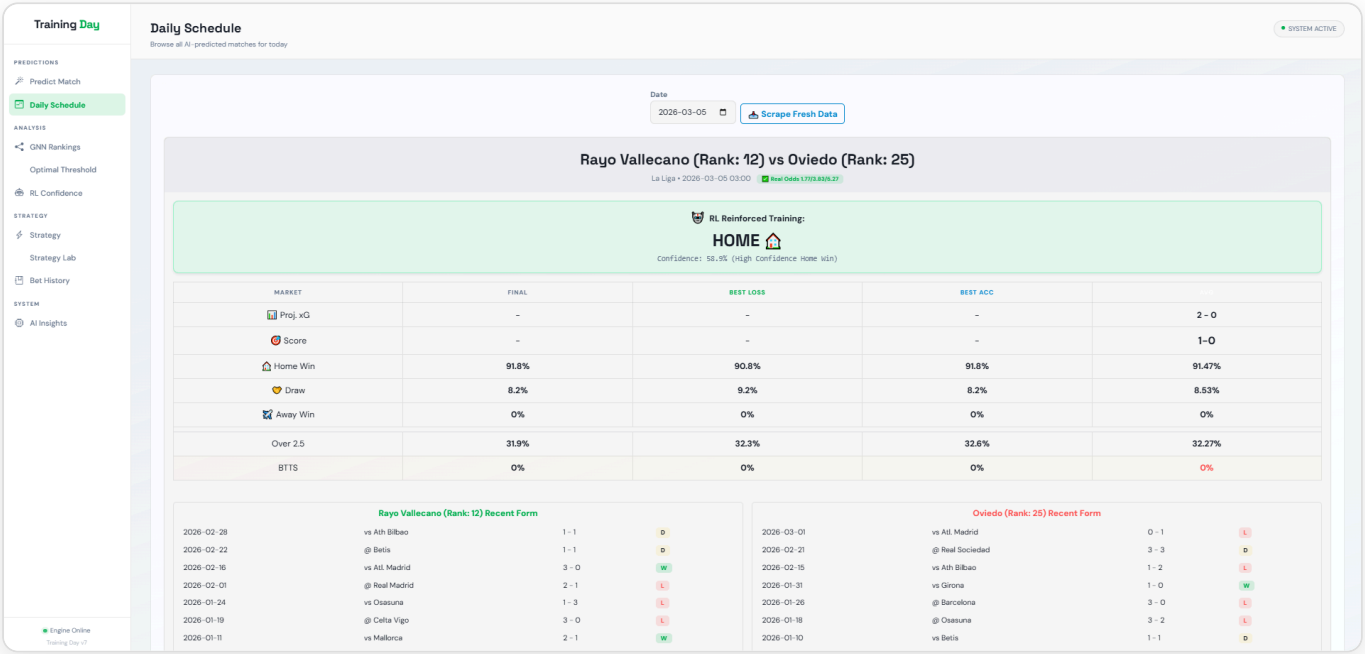
**Step 2** Predictive Ensembling: Triple architecture — TensorFlow DL + XGBoost + LightGBM.

**Step 3** Confidence Optimization: Reinforcement Learning (RL) agent for tuning logical confidence thresholds.

**Step 4** Probabilistic Reasoning: Monte Carlo simulation for calibrated outcome probabilities.

**Step 5** Dynamic Tuning: Adaptive sequence length optimization based on league-specific volatility.

**Step 6** Power Ranking: ELO-based strength metrics and GNN-based dominance analysis.



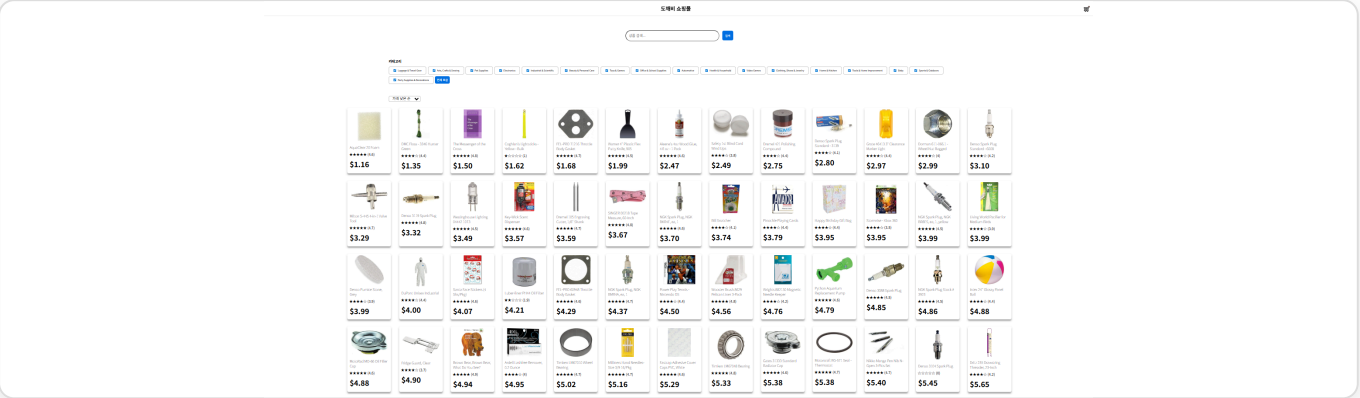
## OTHER PROJECTS

### Goblin Shopping Mall (E-commerce SPA)

Hyunsik Jeon and 3 others

*React · JavaScript · CSS · Vercel*

A fully-featured e-commerce platform with 18+ category filters and real-time search. Focused on intuitive UX and high-performance frontend state management.



### Crazy Arcade Web Replica

Hyunsik Jeon and 3 others

*React · FastAPI · WebSocket · MySQL*

A real-time multiplayer game synchronization project using WebSockets. Implemented complex physics and room-state management for smooth gameplay.

### My Diary (Emotion Log App)

(Personal Use for Education)

*React · Context API · localStorage*

A personal journal application focusing on frontend state persistence and custom UI interactions using browser's local storage.